

STAT 220B Midterm Solutions

Regular Midterm and Makeup Midterm

Lecture plan

Today we use the two midterms as a review lecture on four recurring themes:

1. Sufficiency and Rao–Blackwellization
2. Bayes rules and posterior expected loss
3. Risk, admissibility, and domination
4. Proper versus improper priors

The goal is not just to check answers. The goal is to see the reusable argument behind each solution.

Big picture: what these exams were testing

The regular midterm focused on:

- ▶ exponential-family sufficiency;
- ▶ Rao–Blackwell improvement;
- ▶ Bayes rules for Bernoulli and Poisson models;
- ▶ Jeffreys priors and generalized Bayes rules;
- ▶ admissibility of proper Bayes rules.

The makeup midterm tested the same conceptual backbone through:

- ▶ finite decision problems and risk sets;
- ▶ expected utility and risk aversion;
- ▶ Rao–Blackwellization for Bernoulli variance estimation;
- ▶ predictive Bayes rules and admissibility.

Regular Midterm

Regular Midterm Problem 1

Sufficiency and Rao–Blackwell for exponential data

Setup:

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Exp}(\theta), \quad f(x \mid \theta) = \theta e^{-\theta x}, \quad x > 0.$$

We estimate

$$g(\theta) = \frac{1}{\theta}$$

under squared-error loss.

The statistic is

$$T = \sum_{i=1}^n X_i, \quad \bar{X} = T/n.$$

Problem 1(a): sufficiency

The joint density is

$$f(x_1, \dots, x_n \mid \theta) = \prod_{i=1}^n \theta e^{-\theta x_i} = \theta^n \exp \left\{ -\theta \sum_{i=1}^n x_i \right\} \prod_{i=1}^n \mathbf{1}\{x_i > 0\}.$$

Write this as

$$f(x \mid \theta) = g_{\theta}(T(x))h(x),$$

where

$$T(x) = \sum_i x_i, \quad g_{\theta}(t) = \theta^n e^{-\theta t}.$$

So, by the factorization theorem, T is sufficient for θ .

Problem 1(a): distribution of T

A sum of n independent exponential random variables with rate θ is Gamma with shape n and rate θ :

$$T \sim \text{Gamma}(n, \theta), \quad f_T(t \mid \theta) = \frac{\theta^n}{(n-1)!} t^{n-1} e^{-\theta t}, \quad t > 0.$$

Teaching point.

Do not just say “exponential family, hence sufficient.” You need to show the factorization and identify exactly what part depends on θ .

Problem 1(b): risk of the crude estimator

The crude estimator is

$$\delta_0(X) = X_1.$$

Since $X_1 \sim \text{Exp}(\theta)$,

$$\mathbb{E}_\theta[X_1] = \frac{1}{\theta}, \quad \text{Var}_\theta(X_1) = \frac{1}{\theta^2}.$$

Thus δ_0 is unbiased for $1/\theta$.

Under squared-error loss, unbiasedness gives

$$R(\theta, \delta_0) = \mathbb{E}_\theta [(X_1 - 1/\theta)^2] = \text{Var}_\theta(X_1) = \frac{1}{\theta^2}.$$

Problem 1(c): Rao–Blackwellized estimator

We compute

$$\delta_1(T) = \mathbb{E}[X_1 \mid T].$$

By exchangeability,

$$\mathbb{E}[X_1 \mid T] = \mathbb{E}[X_2 \mid T] = \cdots = \mathbb{E}[X_n \mid T].$$

Therefore

$$T = \mathbb{E}[T \mid T] = \sum_{i=1}^n \mathbb{E}[X_i \mid T] = n \mathbb{E}[X_1 \mid T].$$

So

$$\delta_1(T) = \frac{T}{n} = \bar{X}.$$

Problem 1(d): risk improvement

Since $\bar{X}$ has mean $1/\theta$ and variance $1/(n\theta^2)$,

$$R(\theta, \delta_1) = \text{Var}_\theta(\bar{X}) = \frac{1}{n\theta^2}.$$

Compare:

$$R(\theta, \delta_0) - R(\theta, \delta_1) = \frac{1}{\theta^2} - \frac{1}{n\theta^2} = \frac{n-1}{n\theta^2} > 0$$

for every $n \geq 2$ and $\theta > 0$.

Message. Rao–Blackwellization replaces a noisy unbiased estimator by its conditional expectation given a sufficient statistic. Here it is literally averaging away irrelevant randomness.

Regular Midterm Problem 2

Bayes rules in the Bernoulli model

Setup:

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta), \quad \theta \in (0, 1),$$

with prior

$$\theta \sim \text{Beta}(\alpha, \beta), \quad \alpha, \beta > 0.$$

Let

$$T = \sum_{i=1}^n X_i.$$

Problem 2(a): posterior

The likelihood based on $T = t$ is

$$P(T = t \mid \theta) = \binom{n}{t} \theta^t (1 - \theta)^{n-t}.$$

Dropping constants that do not involve θ ,

$$L(\theta; t) \propto \theta^t (1 - \theta)^{n-t}.$$

Multiplying by the prior density,

$$\pi(\theta \mid T = t) \propto \theta^{\alpha+t-1} (1 - \theta)^{\beta+n-t-1}.$$

Thus

$$\theta \mid T = t \sim \text{Beta}(\alpha + t, \beta + n - t).$$

The posterior is proper because

$$\alpha + t > 0, \quad \beta + n - t > 0,$$

for $\alpha, \beta > 0$ and $0 \leq t \leq n$.

Problem 2(b): squared-error Bayes rule

Under squared-error loss,

$$\rho(a \mid t) = \mathbb{E}[(\theta - a)^2 \mid T = t] = \text{Var}(\theta \mid T = t) + \{\mathbb{E}(\theta \mid T = t) - a\}^2.$$

The first term is constant in a .

The second term is minimized uniquely at the posterior mean.

Therefore

$$\delta_{\pi}(t) = \mathbb{E}[\theta \mid T = t] = \frac{\alpha + t}{\alpha + \beta + n}.$$

Problem 2(c): Bayes test

We test

$$H_0 : \theta \leq 1/2 \quad \text{versus} \quad H_1 : \theta > 1/2.$$

Actions:

- ▶ a_0 : accept H_0 ;
- ▶ a_1 : reject H_0 .

Loss:

$$L(\theta, a_0) = \mathbf{1}\{\theta > 1/2\}, \quad L(\theta, a_1) = \mathbf{1}\{\theta \leq 1/2\}.$$

Let

$$p(t) = P(\theta \leq 1/2 \mid T = t).$$

Problem 2(c): posterior risks

For action a_0 ,

$$\rho(a_0 \mid t) = P(\theta > 1/2 \mid T = t) = 1 - p(t).$$

For action a_1 ,

$$\rho(a_1 \mid t) = P(\theta \leq 1/2 \mid T = t) = p(t).$$

Choose the action with smaller posterior expected loss:

a_0 if $p(t) > 1/2$,
a_1 if $p(t) < 1/2$.

Tie if $p(t) = 1/2$.

Regular Midterm Problem 3

Jeffreys prior for the Poisson model

Setup:

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Poisson}(\lambda), \quad \lambda > 0.$$

Let

$$T = \sum_{i=1}^n X_i.$$

Squared-error loss:

$$L(\lambda, a) = (a - \lambda)^2.$$

Problem 3(a): Fisher information

For one observation,

$$\log f(x \mid \lambda) = -\lambda + x \log \lambda - \log(x!).$$

Then

$$\frac{\partial}{\partial \lambda} \log f(x \mid \lambda) = -1 + \frac{x}{\lambda},$$

and

$$\frac{\partial^2}{\partial \lambda^2} \log f(x \mid \lambda) = -\frac{x}{\lambda^2}.$$

Thus

$$I(\lambda) = -\mathbb{E}_{\lambda} \left[\frac{\partial^2}{\partial \lambda^2} \log f(X \mid \lambda) \right] = \frac{\mathbb{E}_{\lambda}[X]}{\lambda^2} = \frac{1}{\lambda}.$$

Problem 3(a): Jeffreys prior is improper

Jeffreys prior is

$$\pi_J(\lambda) \propto I(\lambda)^{1/2} = \lambda^{-1/2}.$$

But

$$\int_0^\infty \lambda^{-1/2} d\lambda = \lim_{M \rightarrow \infty} \int_0^M \lambda^{-1/2} d\lambda = \lim_{M \rightarrow \infty} 2\sqrt{M} = \infty.$$

So π_J is **improper**.

Teaching point.

Improper prior does not automatically mean improper posterior. But it does mean admissibility arguments based on proper Bayes rules cannot be used directly.

Problem 3(b): likelihood for T

Since sums of independent Poissons are Poisson,

$$T \mid \lambda \sim \text{Poisson}(n\lambda).$$

Thus

$$P(T = t \mid \lambda) = \frac{e^{-n\lambda} (n\lambda)^t}{t!} \propto \lambda^t e^{-n\lambda},$$

where the proportionality drops $n^t/t!$, which does not depend on λ .

Combining with the Jeffreys prior $\pi_J(\lambda) \propto \lambda^{-1/2}$ gives

$$\pi_J(\lambda \mid T = t) \propto \lambda^{-1/2} \lambda^t e^{-n\lambda} = \lambda^{t-1/2} e^{-n\lambda} = \lambda^{(t+1/2)-1} e^{-n\lambda}.$$

Problem 3(b): identify and check propriety

The posterior kernel is

$$\pi_J(\lambda \mid T = t) \propto \lambda^{(t+1/2)-1} e^{-n\lambda}.$$

Therefore

$$\lambda \mid T = t \sim \text{Gamma}(t + 1/2, n)$$

in the shape-rate parameterization.

This formal posterior is proper for every $t = 0, 1, 2, \dots$ because the shape parameter is positive:

$$t + 1/2 > 0.$$

Indeed,

$$\int_0^\infty \lambda^{t-1/2} e^{-n\lambda} d\lambda = \frac{\Gamma(t + 1/2)}{n^{t+1/2}} < \infty.$$

Problem 3(c): generalized Bayes estimator

Under squared-error loss, the Bayes action is the posterior mean.

For

$$\lambda \mid T = t \sim \text{Gamma}(t + 1/2, n),$$

the posterior mean is shape divided by rate:

$$\delta_J(T) = \frac{T + 1/2}{n}.$$

This is a **generalized Bayes estimator** because the prior is improper.

Problem 3(d): risk of the generalized Bayes rule

Since $T \sim \text{Poisson}(n\lambda)$,

$$\mathbb{E}_{\lambda}[T] = n\lambda, \quad \text{Var}_{\lambda}(T) = n\lambda.$$

For

$$\delta_J(T) = \frac{T + 1/2}{n},$$

we have

$$\mathbb{E}_{\lambda}[\delta_J] = \lambda + \frac{1}{2n}, \quad \text{Var}_{\lambda}(\delta_J) = \frac{\lambda}{n}.$$

Thus

$$R(\lambda, \delta_J) = \frac{\lambda}{n} + \frac{1}{4n^2}.$$

Problem 3(d): compare with MLE

The MLE is

$$\delta_{\text{MLE}}(T) = \frac{T}{n}.$$

It is unbiased and has variance λ/n , so

$$R(\lambda, \delta_{\text{MLE}}) = \frac{\lambda}{n}.$$

Therefore

$$R(\lambda, \delta_J) - R(\lambda, \delta_{\text{MLE}}) = \frac{1}{4n^2} > 0.$$

So the MLE strictly dominates δ_J for every $\lambda > 0$.

δ_J is inadmissible.

Regular Midterm Problem 4

Admissibility of a proper Bayes rule

Setup:

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Poisson}(\lambda), \quad \lambda > 0,$$

with prior

$$\lambda \sim \text{Gamma}(\alpha, \beta), \quad \alpha, \beta > 0.$$

Use squared-error loss.

Problem 4(a): posterior

The likelihood based on $T = t$ is

$$L(\lambda; t) \propto \lambda^t e^{-n\lambda}.$$

The prior is

$$\pi(\lambda) \propto \lambda^{\alpha-1} e^{-\beta\lambda}.$$

Therefore

$$\pi(\lambda \mid T = t) \propto \lambda^{\alpha+t-1} e^{-(\beta+n)\lambda}.$$

Thus

$$\lambda \mid T = t \sim \text{Gamma}(\alpha + t, \beta + n).$$

Problem 4(b): Bayes rule

The squared-error Bayes rule is the posterior mean:

$$\delta_{\alpha, \beta}(T) = \frac{\alpha + T}{\beta + n}.$$

This is a proper Bayes rule because the prior is proper.

Problem 4(c): frequentist risk

For

$$\delta_{\alpha,\beta}(T) = \frac{\alpha + T}{\beta + n},$$

variance:

$$\text{Var}_{\lambda}(\delta_{\alpha,\beta}) = \frac{\text{Var}_{\lambda}(T)}{(\beta + n)^2} = \frac{n\lambda}{(\beta + n)^2}.$$

bias:

$$\mathbb{E}_{\lambda}[\delta_{\alpha,\beta}] - \lambda = \frac{\alpha + n\lambda}{\beta + n} - \lambda = \frac{\alpha - \beta\lambda}{\beta + n}.$$

Therefore

$$R(\lambda, \delta_{\alpha,\beta}) = \frac{n\lambda + (\alpha - \beta\lambda)^2}{(\beta + n)^2}.$$

Problem 4(d): why uniqueness matters

At fixed $T = t$,

$$\rho(a \mid t) = \mathbb{E}[(\lambda - a)^2 \mid T = t] = \text{Var}(\lambda \mid T = t) + \{a^*(t) - a\}^2,$$

where

$$a^*(t) = \mathbb{E}[\lambda \mid T = t] = \frac{\alpha + t}{\beta + n}.$$

Equivalently, as a function of a ,

$$\rho(a \mid t) = a^2 - 2a^*(t)a + \{\text{Var}(\lambda \mid T = t) + a^*(t)^2\}.$$

The coefficient of a^2 is positive, so this is strictly convex and has a unique global minimizer at $a = a^*(t)$.

Problem 4(d): uniqueness of the Bayes rule

For any rule δ' , the tower property gives

$$r(\pi, \delta') = \mathbb{E}_T[\rho(\delta'(T) \mid T)] \geq \mathbb{E}_T[\rho(a^*(T) \mid T)] = r(\pi, \delta_{\alpha, \beta}).$$

Equality holds only if

$$\delta'(T) = a^*(T)$$

almost surely under the marginal distribution of T .

The marginal distribution of T gives positive probability to every $t = 0, 1, 2, \dots$ because $P(T = t \mid \lambda) > 0$ for every $\lambda > 0$, and the Gamma prior has full support on $(0, \infty)$.

Therefore almost-sure equality forces equality at every nonnegative integer t , so $\delta_{\alpha, \beta}$ is the unique Bayes rule.

Problem 4(d): finite Bayes risk and admissibility

Using the risk expression,

$$R(\lambda, \delta_{\alpha, \beta}) = \frac{\beta^2 \lambda^2 + (n - 2\alpha\beta)\lambda + \alpha^2}{(\beta + n)^2}.$$

By Tonelli's theorem, since risk is nonnegative,

$$r(\pi, \delta_{\alpha, \beta}) = \mathbb{E}_{\pi}[R(\lambda, \delta_{\alpha, \beta})].$$

For $\lambda \sim \text{Gamma}(\alpha, \beta)$,

$$\mathbb{E}_{\pi}[\lambda] = \frac{\alpha}{\beta}, \quad \mathbb{E}_{\pi}[\lambda^2] = \frac{\alpha(\alpha + 1)}{\beta^2}.$$

Problem 4(d): Bayes risk calculation

Therefore

$$\begin{aligned}r(\pi, \delta_{\alpha, \beta}) &= \frac{\beta^2 \mathbb{E}[\lambda^2] + (n - 2\alpha\beta) \mathbb{E}[\lambda] + \alpha^2}{(\beta + n)^2} \\&= \frac{\alpha(\alpha + 1) + (n - 2\alpha\beta)\alpha/\beta + \alpha^2}{(\beta + n)^2} \\&= \frac{\alpha + \alpha n/\beta}{(\beta + n)^2} \\&= \frac{\alpha}{\beta(\beta + n)} < \infty.\end{aligned}$$

Thus, by the unique proper Bayes rule admissibility theorem,

$\delta_{\alpha, \beta}$ is admissible.

Makeup Midterm

Makeup Problem 1

Risk sets, admissibility, and randomization

Finite decision problem:

$$\Theta = \{\theta_0, \theta_1\}, \quad \mathcal{A} = \{a_0, a_1\}, \quad \mathcal{X} = \{1, 2, 3\}.$$

Loss matrix:

	a_0	a_1
θ_0	0	2
θ_1	4	0

Decision rules are labeled δ_{abc} , where $a, b, c \in \{0, 1\}$ are the actions at $X = 1, 2, 3$.

The sampling distributions are

$$P_{\theta_0}(1) = 1/2, \quad P_{\theta_0}(2) = 3/10, \quad P_{\theta_0}(3) = 1/5,$$

and

$$P_{\theta_1}(1) = 1/10, \quad P_{\theta_1}(2) = 1/5, \quad P_{\theta_1}(3) = 7/10.$$

Makeup Problem 1(a): risk table

Use

$$R(\theta_i, \delta) = \sum_{x=1}^3 L(\theta_i, \delta(x)) P_{\theta_i}(X = x).$$

For example,

$$R(\theta_0, \delta_{011}) = 0 \cdot \frac{1}{2} + 2 \cdot \frac{3}{10} + 2 \cdot \frac{1}{5} = 1,$$

and

$$R(\theta_1, \delta_{011}) = 4 \cdot \frac{1}{10} + 0 + 0 = \frac{2}{5}.$$

The risk vectors are:

Rule	$R(\theta_0, \cdot)$	$R(\theta_1, \cdot)$
δ_{000}	0	4
δ_{001}	$2/5$	$6/5$
δ_{010}	$3/5$	$16/5$
δ_{011}	1	$2/5$
δ_{100}	1	$18/5$
δ_{101}	$7/5$	$4/5$
δ_{110}	$8/5$	$14/5$
δ_{111}	2	0

Makeup Problem 1(b): admissibility

A rule is inadmissible if another rule has no larger risk at both states and strictly smaller risk at at least one state.

Dominations:

$$\delta_{010} = (3/5, 16/5) \text{ is dominated by } \delta_{001} = (2/5, 6/5).$$

$$\delta_{100} = (1, 18/5) \text{ is dominated by } \delta_{011} = (1, 2/5).$$

$$\delta_{101} = (7/5, 4/5), \quad \delta_{110} = (8/5, 14/5)$$

are also dominated by $\delta_{011} = (1, 2/5)$.

Thus the admissible nonrandomized rules are

$$\delta_{000}, \delta_{001}, \delta_{011}, \delta_{111}.$$

Makeup Problem 1(c): Bayes rule thresholds

For prior $\pi_0 = P(\theta_0)$,

$$\rho(a_0 | x) = 4P(\theta_1 | x), \quad \rho(a_1 | x) = 2P(\theta_0 | x).$$

Action a_0 is preferred iff

$$4P(\theta_1 | x) < 2P(\theta_0 | x) \iff P(\theta_0 | x) > 2/3.$$

After applying Bayes' rule, strict preference becomes

$$\pi_0 > \pi^*(x), \quad \pi^*(x) = \frac{2P_{\theta_1}(x)}{P_{\theta_0}(x) + 2P_{\theta_1}(x)}.$$

At equality, both actions are Bayes. This is why the final prior ranges below use closed intervals.

Makeup Problem 1(c): threshold values

The three thresholds are

$$\pi^*(1) = \frac{2}{7}, \quad \pi^*(2) = \frac{4}{7}, \quad \pi^*(3) = \frac{7}{8}.$$

Therefore:

δ_{011} is Bayes for $\pi_0 \in [2/7, 4/7]$.

δ_{001} is Bayes for $\pi_0 \in [4/7, 7/8]$.

δ_{010} is Bayes for no prior.

The last one fails because it would require both $\pi_0 \leq 4/7$ and $\pi_0 \geq 7/8$.

Makeup Problem 1(d): Bayes rules tile the prior interval

The admissible rules tile $[0, 1]$:

$$\delta_{111} : [0, 2/7], \quad \delta_{011} : [2/7, 4/7],$$

$$\delta_{001} : [4/7, 7/8], \quad \delta_{000} : [7/8, 1].$$

So, in this finite problem, the nonrandomized admissible rules coincide with the nonrandomized Bayes rules for some prior.

Takeaway.

Risk geometry and Bayes optimality are two views of the same object in finite decision problems.

Makeup Problem 1(e): equalizer randomization

Consider

$$\delta^* = \lambda \delta_{001} + (1 - \lambda) \delta_{011}.$$

Its risk vector is

$$\lambda(2/5, 6/5) + (1 - \lambda)(1, 2/5).$$

Set the two components equal:

$$\lambda \frac{2}{5} + (1 - \lambda)1 = \lambda \frac{6}{5} + (1 - \lambda) \frac{2}{5}.$$

Solving gives

$$\lambda^* = \frac{3}{7}.$$

The common risk is

$$\frac{26}{35}.$$

Makeup Problem 2

Utility theory and risk aversion

Utility:

$$U(w) = \log w.$$

Initial wealth:

$$w_0 = 100.$$

Gamble:

$$W = 50 \quad \text{or} \quad W = 160,$$

each with probability $1/2$.

Makeup Problem 2(a): expected wealth and utility

Expected wealth:

$$\mathbb{E}[W] = \frac{1}{2}(50) + \frac{1}{2}(160) = 105.$$

Expected utility:

$$\mathbb{E}[U(W)] = \frac{1}{2} \log 50 + \frac{1}{2} \log 160 = \frac{1}{2} \log(8000) = \log \sqrt{8000}.$$

Makeup Problem 2(b): accept or reject?

Without the gamble,

$$U(w_0) = \log 100.$$

With the gamble,

$$\mathbb{E}[U(W)] = \log \sqrt{8000}.$$

Since

$$\sqrt{8000} \approx 89.44 < 100,$$

we have

$$\mathbb{E}[U(W)] < U(100).$$

So the decision maker rejects the gamble.

Makeup Problem 2(c)–(d): CE and risk premium

The certainty equivalent solves

$$\log(\text{CE}) = \mathbb{E}[\log W] = \log \sqrt{8000}.$$

Thus

$$\text{CE} = \sqrt{8000} = 40\sqrt{5}.$$

The risk premium is

$$\text{RP} = \mathbb{E}[W] - \text{CE} = 105 - 40\sqrt{5} \approx 15.56.$$

Makeup Problem 2(e): why risk premium is positive

Because $U(w) = \log w$ is strictly concave,

$$\mathbb{E}[U(W)] < U(\mathbb{E}[W])$$

for any non-degenerate gamble.

Since U is increasing,

$$U(\text{CE}) = \mathbb{E}[U(W)] < U(\mathbb{E}[W]) \implies \text{CE} < \mathbb{E}[W].$$

Therefore

$$\text{RP} = \mathbb{E}[W] - \text{CE} > 0.$$

Translation. A risk-averse person discounts a risky payoff relative to its arithmetic mean.

Makeup Problem 3

Rao–Blackwell, strict convexity, and inadmissibility

Setup:

$$Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta), \quad \theta \in (0, 1),$$

and the target is

$$g(\theta) = \theta(1 - \theta) = \text{Var}(Y_1).$$

Start with

$$\delta_0(Y_1, Y_2) = \frac{(Y_1 - Y_2)^2}{2}.$$

Makeup Problem 3(a): sufficiency

The joint pmf is

$$P(Y_1 = y_1, \dots, Y_n = y_n \mid \theta) = \theta^{\sum_i y_i} (1 - \theta)^{n - \sum_i y_i}.$$

It depends on the data only through

$$T = \sum_{i=1}^n Y_i.$$

Therefore T is sufficient by factorization, and

$$T \sim \text{Binomial}(n, \theta).$$

Makeup Problem 3(b): unbiasedness

Since $Y_i \in \{0, 1\}$,

$$Y_i^2 = Y_i, \quad \mathbb{E}[Y_i] = \theta.$$

Also, by independence,

$$\mathbb{E}[Y_1 Y_2] = \theta^2.$$

Therefore

$$\mathbb{E}[\delta_0] = \frac{1}{2} \mathbb{E}[(Y_1 - Y_2)^2] = \frac{1}{2} (\theta - 2\theta^2 + \theta) = \theta(1 - \theta).$$

Makeup Problem 3(c): Rao–Blackwellized estimator

Because Y_i is binary,

$$(Y_1 - Y_2)^2 = \mathbf{1}\{Y_1 \neq Y_2\}.$$

So

$$\delta_1(T) = \mathbb{E}[\delta_0 \mid T] = \frac{1}{2} P(Y_1 \neq Y_2 \mid T).$$

Given $T = t$, the vector $(Y_1, \dots, Y_n)$ is uniform over all binary strings with exactly t ones.

Makeup Problem 3(c): counting argument

Among ordered pairs (i, j) with $i \neq j$, the number with unequal values is

$$t(n-t) + (n-t)t = 2t(n-t).$$

There are

$$n(n-1)$$

ordered pairs in total.

By exchangeability,

$$P(Y_1 \neq Y_2 \mid T = t) = \frac{2t(n-t)}{n(n-1)}.$$

Thus

$$\delta_1(T) = \frac{T(n-T)}{n(n-1)}.$$

Makeup Problem 3(d): strict improvement

For $t \in \{1, \dots, n-1\}$,

$$\delta_0 \mid T = t$$

takes value $1/2$ with probability

$$p_t = \frac{2t(n-t)}{n(n-1)}$$

and value 0 otherwise.

For $1 \leq t \leq n-1$,

$$0 < p_t < 1.$$

Therefore

$$\text{Var}(\delta_0 \mid T = t) = \frac{1}{4} p_t (1 - p_t) > 0.$$

For the endpoints $t = 0$ and $t = n$, all observations are equal, so δ_0 is identically zero conditional on $T = t$ and the conditional variance is zero.

Makeup Problem 3(d)–(e): risk decomposition

Both δ_0 and δ_1 are unbiased for $\theta(1 - \theta)$.

Using variance decomposition,

$$\text{Var}(\delta_0) = \mathbb{E}[\text{Var}(\delta_0 \mid T)] + \text{Var}(\mathbb{E}[\delta_0 \mid T]).$$

But

$$\mathbb{E}[\delta_0 \mid T] = \delta_1(T).$$

So

$$R(\theta, \delta_0) - R(\theta, \delta_1) = \mathbb{E}_\theta[\text{Var}(\delta_0 \mid T)] = \sum_{t=0}^n \text{Var}(\delta_0 \mid T = t) P_\theta(T = t).$$

For $\theta \in (0, 1)$, $P_\theta(T = t) > 0$ for all $t = 0, 1, \dots, n$, and at least one interior t has positive conditional variance. Hence the risk improvement is strict for every $\theta \in (0, 1)$.

Therefore

δ_0 is inadmissible.

Makeup Problem 4

Predictive Bayes rule and admissibility

Setup:

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\theta, \sigma^2), \quad \sigma^2 \text{ known,}$$

with prior

$$\theta \sim N(\mu, \tau^2).$$

We predict

$$X_{n+1} \mid \theta \sim N(\theta, \sigma^2)$$

under prediction loss

$$L^*(z, a) = (z - a)^2.$$

Makeup Problem 4(a): reduce prediction to estimation

The induced loss is

$$L(\theta, a) = \mathbb{E}_{\theta}[(X_{n+1} - a)^2].$$

Since

$$\mathbb{E}_{\theta}[X_{n+1}] = \theta, \quad \text{Var}_{\theta}(X_{n+1}) = \sigma^2,$$

we get

$$L(\theta, a) = \sigma^2 + (\theta - a)^2.$$

The constant σ^2 does not affect which action minimizes posterior expected loss.

Makeup Problem 4(b): posterior distribution

We have

$$\bar{X} \mid \theta \sim N\left(\theta, \frac{\sigma^2}{n}\right), \quad \theta \sim N(\mu, \tau^2).$$

The posterior kernel is

$$\pi(\theta \mid \bar{X}) \propto \exp\left[-\frac{1}{2} \left\{ \frac{(\theta - \mu)^2}{\tau^2} + \frac{n(\theta - \bar{X})^2}{\sigma^2} \right\}\right].$$

Collecting the quadratic terms in θ gives precision

$$\frac{1}{\tau^2} + \frac{n}{\sigma^2}$$

and linear coefficient

$$\frac{\mu}{\tau^2} + \frac{n\bar{X}}{\sigma^2}.$$

Thus

$$\theta \mid \bar{X} \sim N(\mu_n, \tau_n^2),$$

where

Makeup Problem 4(c): Bayes prediction rule

The posterior expected prediction loss is

$$\rho(a \mid \bar{X}) = \mathbb{E}[\sigma^2 + (\theta - a)^2 \mid \bar{X}] = \sigma^2 + \tau_n^2 + (\mu_n - a)^2.$$

The only action-dependent part is

$$(\mu_n - a)^2.$$

So the Bayes rule is

$$\delta_\pi(\bar{X}) = \mu_n = \frac{\sigma^2 \mu + n \tau^2 \bar{X}}{\sigma^2 + n \tau^2}.$$

This is the same rule as for estimating θ under squared-error loss.

Makeup Problem 4(d): uniqueness

At every observed $\bar{X}$,

$$\rho(a \mid \bar{X}) = \sigma^2 + \tau_n^2 + (\mu_n - a)^2$$

is a strictly convex quadratic in a .

Therefore the minimizer is unique:

$$a = \mu_n(\bar{X}).$$

The marginal distribution of $\bar{X}$ under the prior is

$$\bar{X} \sim N\left(\mu, \tau^2 + \frac{\sigma^2}{n}\right),$$

which has full support on $\mathbb{R}$.

So equality almost surely forces equality everywhere for continuous rules.

Makeup Problem 4(d): Bayes risk and admissibility

Under the induced loss,

$$R(\theta, \delta_\pi) = \sigma^2 + \text{Var}_\theta(\delta_\pi) + \text{Bias}_\theta(\delta_\pi)^2.$$

The bias is

$$\frac{\sigma^2(\mu - \theta)}{\sigma^2 + n\tau^2},$$

and the variance is

$$\left(\frac{n\tau^2}{\sigma^2 + n\tau^2} \right)^2 \frac{\sigma^2}{n}.$$

Thus the frequentist risk is

$$R(\theta, \delta_\pi) = \sigma^2 + \frac{n\sigma^2\tau^4 + \sigma^4(\mu - \theta)^2}{(\sigma^2 + n\tau^2)^2}.$$

Averaging over $\theta \sim N(\mu, \tau^2)$ gives

$$\begin{aligned} r(\pi, \delta_\pi) &= \sigma^2 + \frac{n\sigma^2\tau^4 + \sigma^4\tau^2}{(\sigma^2 + n\tau^2)^2} \\ &= \sigma^2 + \frac{\sigma^2\tau^2}{\sigma^2 + n\tau^2} \end{aligned}$$

Synthesis

The recurring proof templates

Template 1: Sufficiency.

Show

$$f(x \mid \theta) = g_{\theta}(T(x))h(x).$$

Template 2: Squared-error Bayes rule.

Use

$$\mathbb{E}[(\theta - a)^2 \mid X] = \text{Var}(\theta \mid X) + \{\mathbb{E}(\theta \mid X) - a\}^2.$$

Template 3: Rao–Blackwell improvement.

Use

$$\delta_1 = \mathbb{E}[\delta_0 \mid T]$$

and compare risks through conditional variance.

The recurring admissibility logic

To prove inadmissibility:

Find a rule δ' such that

$$R(\theta, \delta') \leq R(\theta, \delta)$$

for all θ , with strict inequality somewhere.

Examples:

- ▶ Jeffreys Poisson estimator dominated by the MLE;
- ▶ crude Bernoulli variance estimator dominated by its Rao–Blackwellization.

To prove admissibility:

Show the rule is:

1. Bayes under a proper prior;
2. unique;
3. finite Bayes risk.

Then apply the unique Bayes rule admissibility theorem.

What students should remember

The exam was not about memorizing special distributions. It was about recognizing four moves:

1. **Compress the data** using sufficiency.
2. **Average out noise** using Rao–Blackwellization.
3. **Minimize posterior risk** action-by-action.
4. **Compare risks** to decide admissibility.

If those four moves are automatic, most of the calculations become bookkeeping.